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Hsieh et al.

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(54) **AXIAL-FLOW HEAT-DISSIPATION FAN**

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F04D 19/00 (2006.01)
F04D 29/32 (2006.01)
F04D 29/52 (2006.01)

(52) **U.S. Cl.**

CPC ***F04D 29/522*** (2013.01); ***F04D 19/002*** (2013.01); ***F04D 29/325*** (2013.01)

(58) **Field of Classification Search**

CPC F04D 29/522; F04D 19/002; F04D 29/667; F04D 29/68

See application file for complete search history.

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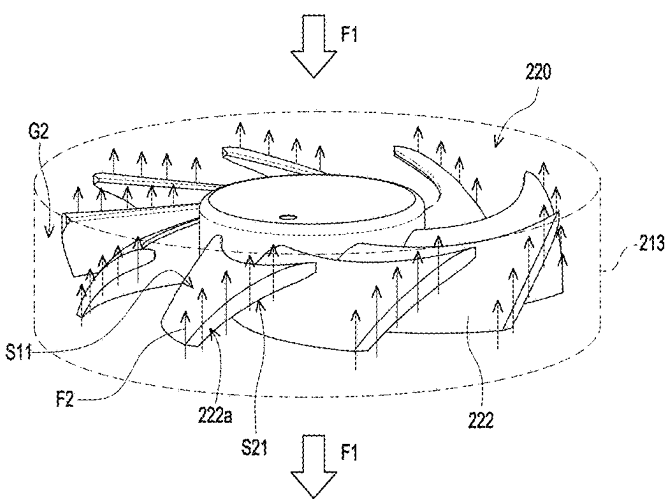
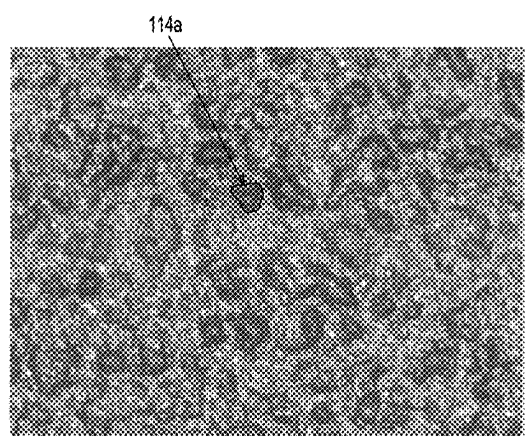
(57) **ABSTRACT**

An axial-flow heat-dissipation fan including a frame and a blade wheel is provided. The frame has an inlet, an outlet, and an inner wall connected between the inlet and the outlet. The inner wall surrounding the blade wheel has at least one rough region. The blade wheel is rotatably disposed in the frame and located between the inlet and the outlet, and an air flows into the frame via the inlet and flows out of the frame via the outlet by rotation of the blade wheel. A gap exists between a blade end of the blade wheel and the inner wall. A laminar flow is generated at the gap when the blade wheel is rotating and the blade end passes through the rough region so as to prevent a backflow generated at the gap, wherein a flowing direction of the backflow is opposite to a flowing direction of the air flow.

6 Claims, 4 Drawing Sheets





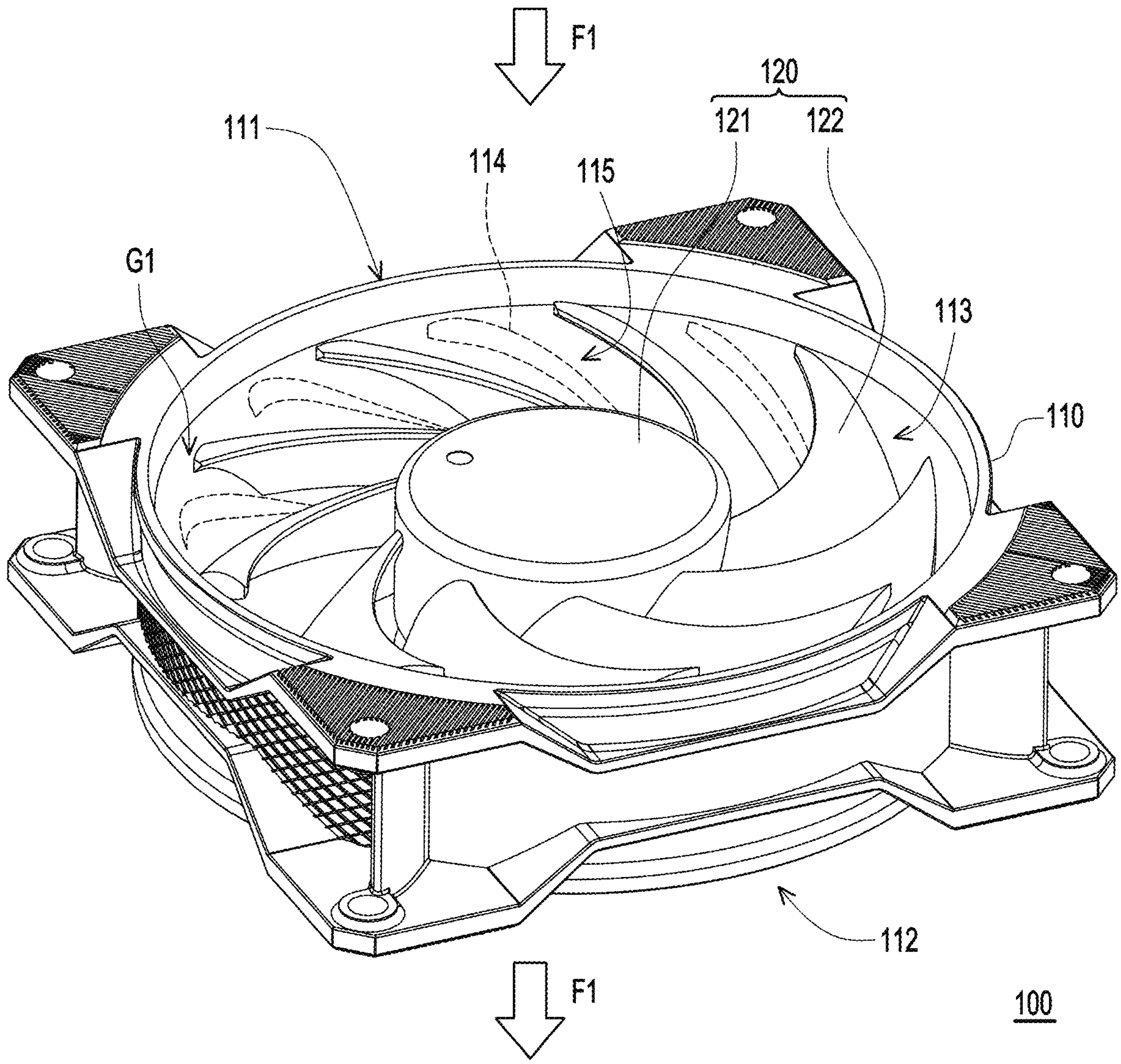


FIG. 1

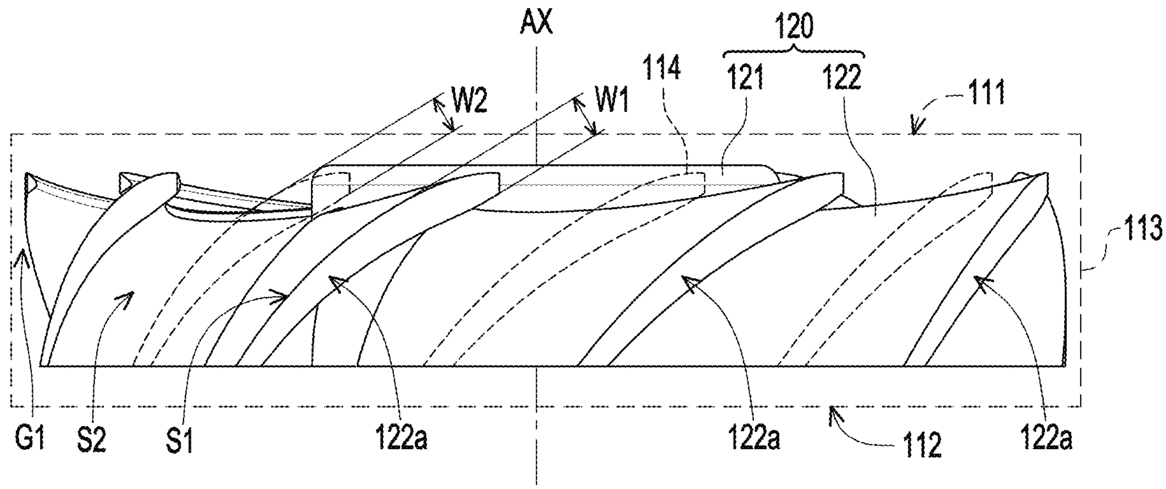

AX
120
114
121
122
W2
W1
111
113
G1
S2
S1
122a
122a
112
122a

FIG. 2

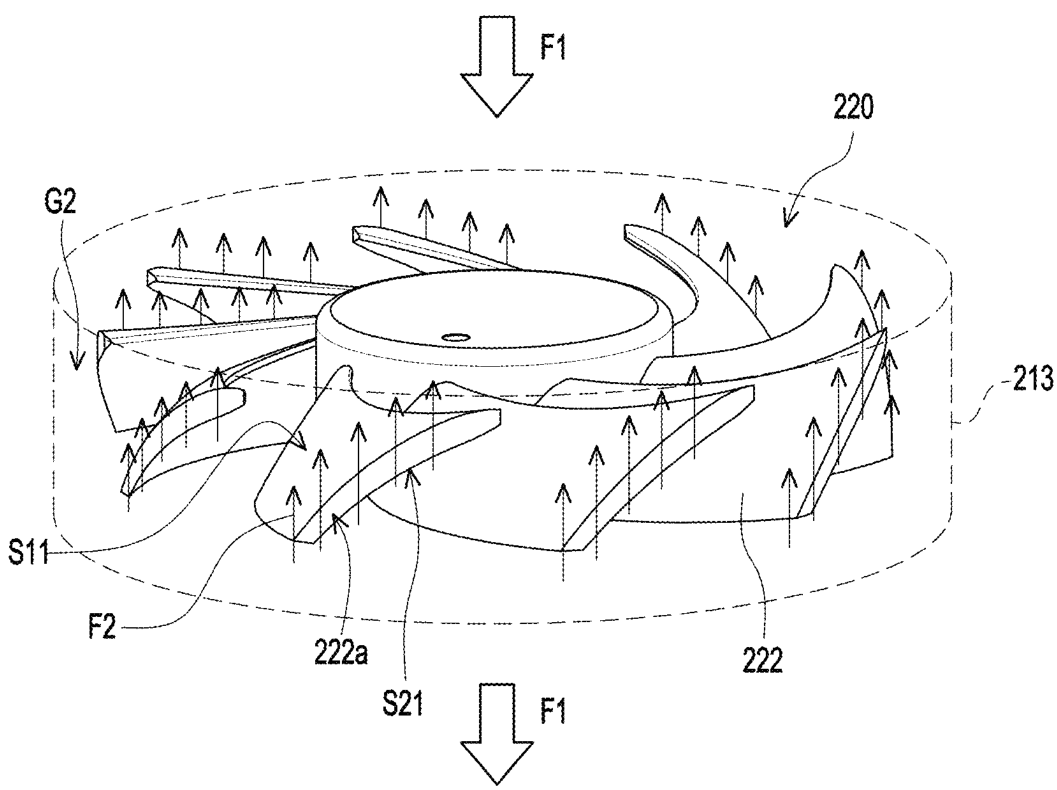

F1
220
G2
213
S11
F2
222a
S21
222
F1

FIG. 3

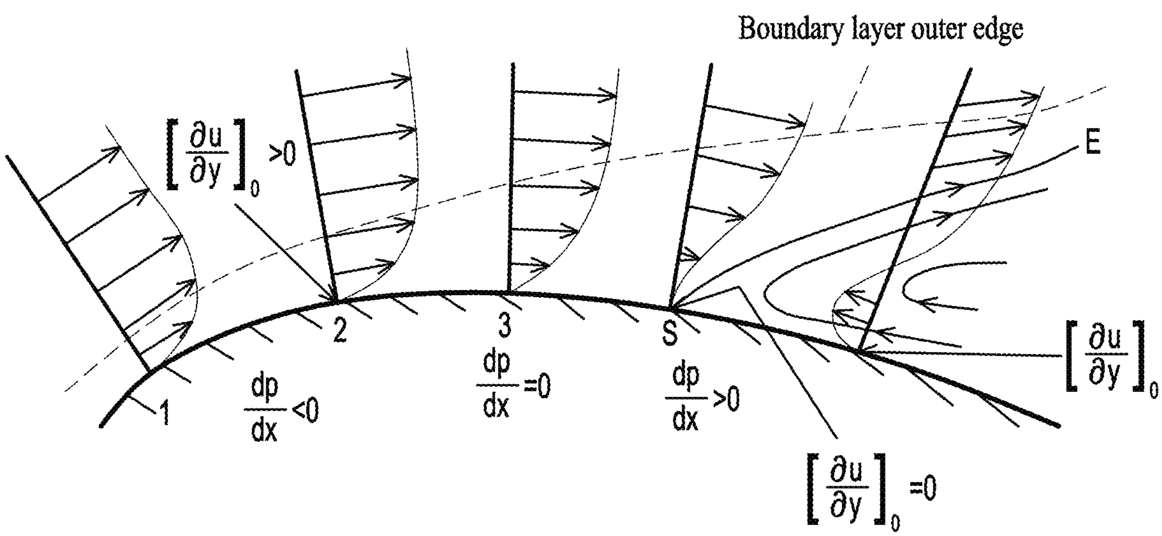

Boundary layer outer edge
E
$\left[\frac{\partial u}{\partial y}\right]_0 > 0$
1
2
3
S
$\frac{dp}{dx} < 0$
$\frac{dp}{dx} = 0$
$\frac{dp}{dx} > 0$
$\left[\frac{\partial u}{\partial y}\right]_0$
$\left[\frac{\partial u}{\partial y}\right]_0 = 0$

FIG. 4

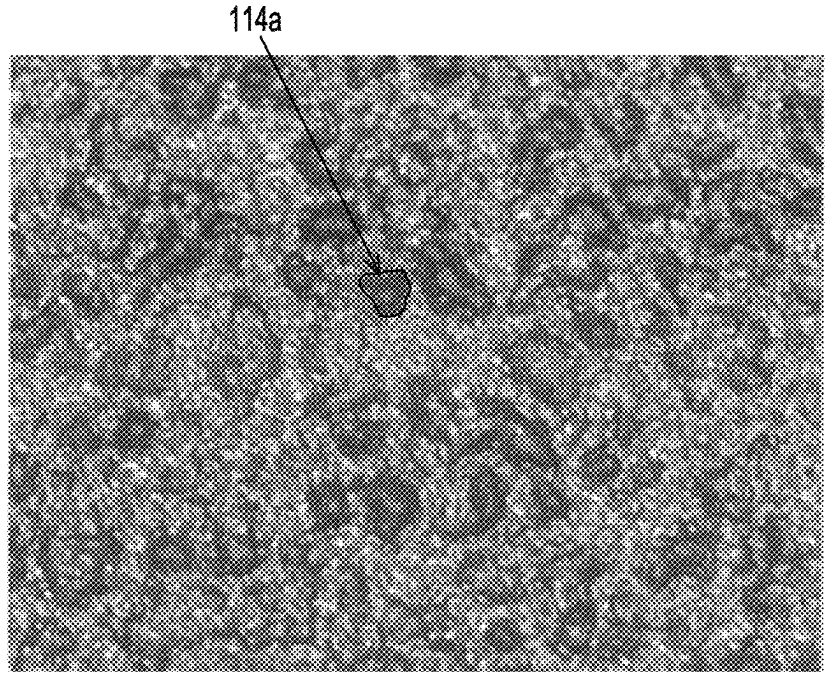


FIG. 5

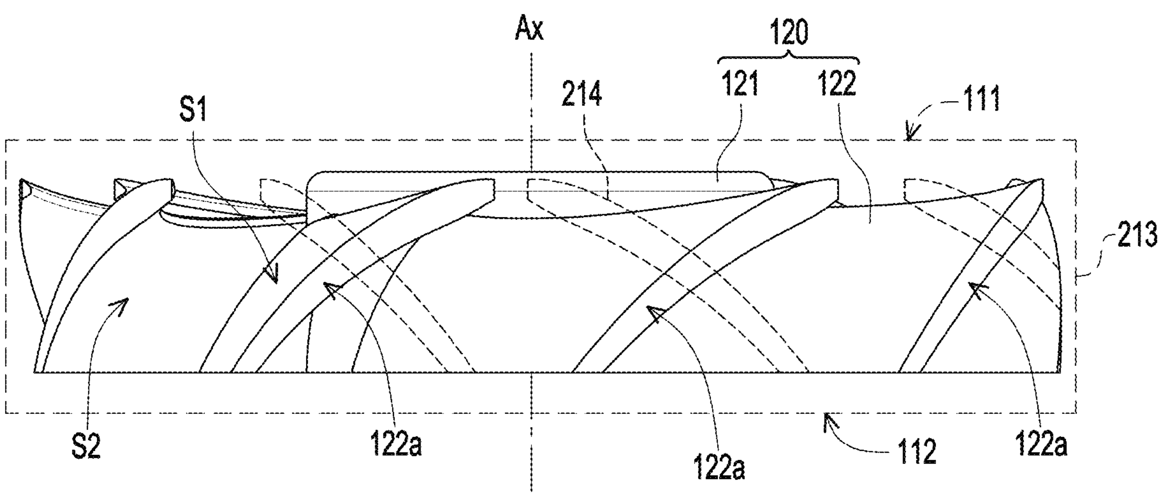


FIG. 6

AXIAL-FLOW HEAT-DISSIPATION FAN

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of Taiwan application serial no. 112134706 filed on Sep. 12, 2023. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

The disclosure relates to a heat-dissipation fan, and in particular relates to an axial-flow heat-dissipation fan.

Description of Related Art

Axial-flow fans have a simple structure and have air supply characteristics of high air volume and low static pressure. Therefore, they are widely used as heat-dissipation fans or ventilation fans for personal computers and servers. Currently, in order to improve the air supply characteristics of axial-flow fans and reduce noise for other optimization purposes, various designs and tests are often conducted on the number of fan blades, the structure of fan blades, or the structure where the air flow passes.

In the above-mentioned adjustment items for the structure where the air flow passes, one of them is to reduce the gap between the fan blades and the frame, thereby reducing the pressure difference between the two opposite surfaces of the blades that causes the backflow of the air flow. However, the above-mentioned gap reduction process, for example, reducing the gap between the blade end and the frame from 1 mm to 0.5 mm, is limited by the precision of the fan component manufacturing and the assembly process. Although it may achieve the desired objective, it results in lower yield rates and increased costs, which is not conducive to mass production.

SUMMARY

An axial-flow heat-dissipation fan is provided in the disclosure, in which backflow is prevented from generating between the blade end and an inner wall of the frame through the rough inner wall of the frame.

The axial-flow heat-dissipation fan of the disclosure is adapted for an electronic device, and includes a frame and a blade wheel. The frame has an inlet, an outlet, and an inner wall connected between the inlet and the outlet. The inner wall surrounding the blade wheel has at least one rough region. The blade wheel is rotatably disposed in the frame and located between the inlet and the outlet, and an air flow generated by rotation of the blade wheel flows into the frame via the inlet and flows out of the frame via the outlet. A gap exists between a blade end of the blade wheel and the inner wall. A laminar flow is generated at the gap when the blade wheel is rotating and the blade end passes through the rough region so as to prevent a backflow generated at the gap. A flowing direction of the backflow is opposite to a flowing direction of the air flow.

Based on the above, for the axial-flow heat-dissipation fan, since at least one rough region is formed on the inner wall of the frame, when the blade wheel rotates and the blade end of the blade wheel passes through the rough region, laminar flow is generated at a gap between the blade end and the inner wall, so that the backflow at the gap is blocked due to the existence of the laminar flow (to prevent the generation of backflow). In this way, compared with the existing method of reducing the gap, which results in poor yield and increased manufacturing costs, the disclosure may effectively prevent backflow with a simple rough structure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a schematic diagram of an axial-flow heat-dissipation fan according to an embodiment of the disclosure.

FIG. **2** is a simple side view of the axial-flow heat-dissipation fan.

FIG. **3** is a simple schematic diagram of the backflow generated by the current axial-flow fan.

FIG. **4** is a simple schematic diagram of a fluid boundary layer.

FIG. **5** is a structural schematic diagram of the rough region.

FIG. **6** is a simple side view of an axial-flow heat-dissipation fan of another embodiment of the disclosure.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

FIG. **1** is a schematic diagram of an axial-flow heat-dissipation fan according to an embodiment of the disclosure. FIG. **2** is a simple side view of the axial-flow heat-dissipation fan. Referring to FIG. **1** and FIG. **2** at the same time, in this embodiment, the axial-flow heat-dissipation fan **100** is adapted for an electronic device, such as a system fan or a CPU heat-dissipation fan installed in a desktop computer, including a frame **110** and a blade wheel **120**. The blade wheel **120** includes a hub **121** and multiple blades **122** surrounding the hub **121**. The frame **110** has an inlet **111**, an outlet **112**, and an inner wall **113**. The inner wall **113** is connected between the inlet **111** and the outlet **112**, in which the inner wall **113** surrounds the blade wheel **120** and has at least one rough region (multiple rough regions **114** are provided in the drawing as an example, and the range of the rough regions is marked with a dashed line). The blade wheel **120** is rotatably disposed in the frame **110** and is located between the inlet **111** and the outlet **112**. The blade wheel **120** is connected through a motor (not shown) and drives the hub **121** to rotate with the axis AX as a reference. When the blades **122** rotate along the axis AX with the hub **121**, the air flow F**1** is generated to flow into the frame **110** via the inlet **111** and flow out of the frame **110** via the outlet **112**.

FIG. **3** is a simple schematic diagram of the backflow generated by the current axial-flow fan. Referring to FIG. **3**, it should be noted here that a gap G**2** exists between the blade end **222***a* of the blade wheel **220** and the inner wall **213** of the frame of the current axial-flow heat-dissipation fan. Therefore, it is inevitable that the blades **222** of the blade wheel **220** have a pressure difference between the air inlet surface S**11** and the air outlet surface S**21**, which causes a backflow F**2** to be formed at the gap G**2** when the blade wheel **220** rotates, and a flowing direction of the backflow F**2** is opposite to a flowing direction of the air flow F**1**.

Accordingly, in view of the aforementioned method of reducing the gap G**2**, which may easily lead to poor precision and increased cost, as shown in FIG. **1** and FIG. **2**, the disclosure starts with the structure of the frame **110**, that is, at least one rough region **114** is formed on the inner wall **113** to generate laminar flow at the gap G**1** when the blade wheel **120** rotates.

FIG. **4** is a simple schematic diagram of a fluid boundary layer. Referring to FIG. **4**, the following is a brief description of the formation principle of laminar flow. Generally speaking, the boundary layer formed between the fluid and the object surface is affected by the roughness of the object surface. As shown on the right side of FIG. **4**, when the fluid pressure increases, the fluid at the inner edge of the boundary layer gradually generates a reverse flow field E due to the viscous resistance of the object surface, causing the fluid to separate from the object surface. This phenomenon of fluid separation is the main cause of laminar flow.

Here, the boundary layer system of equations:

$$u\frac{\partial u}{\partial x}+v\frac{\partial u}{\partial y}\approx U\frac{\partial U}{\partial x}+\frac{\mu}{\rho}\frac{\partial^2 u}{\partial y^2},$$

when the boundary condition is y=0, u=v=0, when y=∞, u=U(x), where u and v represent the velocity components of the fluid in the x and y directions, U(x) represents the flow rate, u represents the dynamic viscosity (dynamic viscosity coefficient), and p represents the fluid density. The direction along the wall of the object is the x-axis, and the direction perpendicular to the wall is the y-axis.

Based on the above principle of boundary layer separation to form laminar flow, the disclosure forms a rough region **114** on the inner wall **113** of the frame **110**, so that laminar flow may be smoothly generated at the gap G**1** when the blade end **122***a* passes through the rough region **114**. In this way, the generation of laminar flow at a particular location essentially serves to block or reduce backflow (such as the aforementioned backflow F**2**) occurring at that location. Here, the blade end **122***a* refers to the side surface adjacent between the air inlet surface S**1** and the air outlet surface S**2** of the blade **122**.

FIG. **5** is a schematic diagram of the structure of the rough region, FIG. **5** is a structural schematic diagram of the rough region, which is shown here as a metallographic image. Referring to FIG. **2** and FIG. **5** at the same time, in this embodiment, the rough region **114** has an etched microstructure and has multiple etching particles **114***a*. The roughness of the rough region **114** is defined by the etching depth of the etched microstructure of 10 μm to 45 μm and the etching particles **114***a* of 15 to 150 per centimeter to ensure the generation of laminar flow. The above-mentioned etching is for the mold that forms the inner wall **113**. By forming the etching pattern on the mold, the corresponding pattern may be successfully formed on the inner wall **113**.

Referring to FIG. **1** and FIG. **2** again, in this embodiment, although the formation of the rough region **114** facilitates the generation of laminar flow, it also reduces the overall flow rate of the air flow **F1** due to increased friction. Accordingly, the disclosure further provides relevant conditions to optimize the characteristics of the rough region **114**. In one embodiment, the inner wall **113** of the frame **110** has multiple rough regions **114**, and the number of the rough regions **114** is greater than or equal to the number of the blades **122** of the blade wheel **120**. Taking FIG. **1** as an example, the blade wheel **120** has nine blades, so the number of rough regions **114** on the inner wall **113** is greater than or equal to nine.

Furthermore, as shown in FIG. **2**, laminar flow is generated due to changes in the air flow at the gap G**1** (by the boundary layer separation), therefore, it substantially depends on the blade end **122***a* and the inner wall **113**, so the outline of the rough region **114** is substantially based on the blade end **122***a*. To be more specific, the rough region **114** is formed by orthogonally projecting the outline of the blade end **122***a* onto the inner wall **113**. Therefore, the width W**2** of the rough region **114** is substantially adjusted based on the width W**1** of the bladed end **122***a*. Based on the above description about the number of rough regions **114**, when the number of rough regions **114** is equal to the number of blades **122**, the minimum width W**2** of the rough region **114** is greater than or equal to the width W**1** of the orthogonal projection outline **122***b* of the blade end **122***a* projected onto the inner wall **113**. When the number of rough regions **114** is greater than the number of blades **122**, the minimum width W**2** of the rough regions **114** is less than the width W**1** of the outline of the blade end **122***a* projected onto the inner wall **113**.

In other words, in this embodiment, the number of rough regions **114** serves as the basis for adjusting the width W**2** of the rough regions **114**. When the number of rough regions **114** is relatively small (e.g., the number of rough regions **114** is equal to the number of blades **122**), the width W**2** of the rough regions **114** may be greater than or equal to the width W**1** of the orthographic projection outline of the blade end **122***a* projected onto the inner wall **113**. Conversely, when the number of rough regions **114** is relatively large (e.g., the number of rough regions **114** is greater than the number of blades **122**), the width W**2** of the rough regions **114** may be appropriately reduced so that the width W**2** of the rough regions **114** is less than the width W**1** of the orthographic projection outline of the blade end **122***a* projected onto the inner wall **113**.

Furthermore, the number and width W**2** of the rough regions **114** may be adjusted based on the area ratio to the inner wall **113**. That is, after determining the total area of these rough regions **114** according to requirements, the width W**2** of the rough regions **114** is adjusted according to the number of blades **122** as a means for optimizing the rough regions **114**. It should be noted that the aforementioned area ratio is based on the area of the inner wall **113** swept by the blade end **122***a* when the blade **122** rotates.

Referring to FIG. **1** and FIG. **2** again, in this embodiment, the rough region **114** is parallel to or consistent with the outline of the blade end **122***a* projected onto the inner wall **113**. This is equivalent to forming rough regions **114** and smooth regions **115** on the inner wall **113**, which are staggered with each other and in a consistent tilt orientation (also consistent with the blade end **122***a*). This allows the blade end **122***a* to generate laminar flow only when passing through the rough region **114**, but may achieve a wider range of laminar flow due to the large overlapping region of the rough region **114** and the orthographic projection outline of the blade end **122***a* on the inner wall **113**.

FIG. **6** is a simple side view of an axial-flow heat-dissipation fan of another embodiment of the disclosure. Referring to FIG. **6**, different from FIG. **2**, the rough region **214** on the inner wall **213** is perpendicular to the orthogonal projection outline of the blade end **122***a* onto the inner wall **213**. This increases the probability that the blade end **122***a* will sweep through the rough region **214**, thereby facilitating the coverage of the region that the blade end **122***a* will sweep over with the least amount of rough region **214**.

Comparing FIG. **2** and FIG. **6** at the same time, the difference lies in the configuration of rough regions **114** and **214**, respectively, in relation to the sweeping of the blade end **122***a*. The rough region **114** may form a larger sweep region (equivalent to covering a gap with a larger range), while the rough region **214** increases the sweep probability by reducing the area of the sweep region (equivalent to more sweeps per unit time). Designers may adjust or mix and match them on the inner wall of the same frame according to actual requirements.

To sum up, in the embodiment of the disclosure, for the axial-flow heat-dissipation fan, since at least one rough region is formed on the inner wall of the frame, when the blade wheel rotates and the blade end of the blade wheel passes through the rough region, laminar flow is generated at a gap between the blade end and the inner wall, so that the backflow at the gap is blocked due to the existence of the laminar flow. In this way, compared with the existing method of reducing the gap, which results in poor yield and increased manufacturing costs, the disclosure may effectively prevent backflow with a simple rough structure.

Furthermore, designers may adjust the number and width of the rough regions according to requirements, or adjust the sweeping relationship between the rough regions and the blade end according to the configuration orientation. In one embodiment, the rough region may be parallel or consistent with the outline of the blade end projected onto the inner wall to generate a wider range of laminar flow through a larger sweep region. In another embodiment, the rough region may be arranged to be perpendicular to the orthogonal projection outline of the blade end projected onto the inner wall, so as to increase the number of times laminar flow is generated per unit time. Designers may adjust and optimize the rough region and its sweeping relationship with the blade end according to requirements.

What is claimed is:

1. An axial-flow heat-dissipation fan, adapted for an electronic device, comprising:

a frame, having an inlet, an outlet, and an inner wall connected between the inlet and the outlet, wherein the inner wall has at least one rough region; and

a blade wheel, rotatably disposed in the frame and located between the inlet and the outlet, the inner wall surrounding the blade wheel, an air flow generated by rotation of the blade wheel flows into the frame via the inlet and flows out of the frame via the outlet, wherein a gap exists between a blade end of the blade wheel and the inner wall, a laminar flow is generated at the gap when the blade wheel is rotating and the blade end passes through the at least one rough region so as to prevent a backflow generated at the gap, a flowing direction of the backflow is opposite to a flowing direction of the air flow, wherein the at least one rough region has an etched microstructure, and a roughness of the at least one rough region is defined by an etching depth of the etched microstructure of 10 μm to 45 μm and etching particles of 15 to 150 per centimeter.

2. The axial-flow heat-dissipation fan according to claim **1**, wherein the at least one rough region is parallel to or consistent with an outline of the blade end orthogonally projected onto the inner wall.

3. The axial-flow heat-dissipation fan according to claim **1**, wherein the at least one rough region is perpendicular to an outline of the blade end orthogonally projected onto the inner wall.

4. The axial-flow heat-dissipation fan according to claim **1**, wherein the at least one rough region of the inner wall comprises a plurality of rough regions, and a number of the rough regions is greater than or equal to a number of blades of the blade wheel.

5. The axial-flow heat-dissipation fan according to claim **1**, wherein the at least one rough region of the inner wall comprises a plurality of rough regions, when a number of the rough regions is equal to a number of blades of the blade wheel, a minimum width of the rough regions is greater than or equal to a width of an outline of the blade end orthogonally projected onto the inner wall.

6. The axial-flow heat-dissipation fan according to claim **1**, wherein the at least one rough region of the inner wall comprises a plurality of rough regions, when a number of the rough regions is greater than a number of blades of the blade wheel, a minimum width of the rough regions is less than a width of an outline of the blade end orthogonally projected onto the inner wall.

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